

Background: Seasonal influenza is a contagious respiratory disease that impacts us every year. The main modes of transmission are through tiny droplets when an infected person coughs, sneezes, or talks within close proximity. Suppose that you are modeling seasonal influenza that spreads between different areas. Figure 1 illustrates the spatial propagation of seasonal influenza across a network. Each node represents a distinct geographical location governed by local SEIR dynamics, while the edges represent the movement of individuals between these regions. The description of the state variables and parameters can be found in Table 1 and Table 2.

Goals:

- Simulate the dynamics of a seasonal influenza model in a network setting.
- Estimate the adjacency matrix from a mixing matrix (cell phone data).
- Using seasonal influenza data, estimate the parameter values for an seasonal influenza outbreak.
- Examine scenarios in which an outbreak of seasonal influenza can be mitigated.
- Understand what parameters have the most influence on disease transmission.

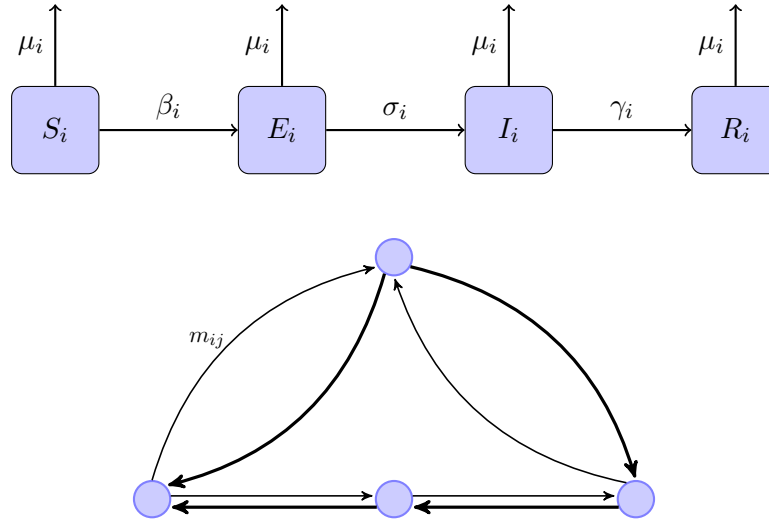


Figure 1: Flow diagram of seasonal influenza model in a network setting.

State Variable	Description
$S_{t,i}$	The number of susceptible individuals in patch i at time t .
$E_{t,i}$	The number of exposed individuals in patch i at time t .
$I_{t,i}$	The number of infected individuals in patch i at time t .
$R_{t,i}$	The number of recovered individuals in patch i at time t .
$N_{t,i}$	The total number of individuals in patch i at time t

Table 1: State variables of influenza model

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Parameter	Description
β_i	Transmission rate in patch i .
σ_i	Transition rate from exposed to infected in patch i .
γ_i	Recovery rate in patch i .
μ_i	Natural death rate in patch i .

Table 2: Parameters of influenza model

Let the parameters $\beta_i = 0.7 \text{ days}^{-1}$, $\sigma_i = 0.1 \text{ days}^{-1}$, $\gamma_i = 0.3 \text{ days}^{-1}$, and $\mu_i = 0.005 \text{ days}^{-1}$ be the same throughout all the patches. Define $S_{0,1} = 98$ individuals, $S_{0,2} = 99$ individuals, $S_{0,3} = 100$ individuals, $E_{0,1} = 1$ individuals, $E_{0,2} = 1$ individuals, $E_{0,3} = 0$ individuals, $I_{0,1} = 1$ individual, $I_{0,2} = 0$ individuals, $I_{0,3} = 0$ individuals, $R_{0,1} = 0$ individuals, $R_{0,2} = 0$ individuals, and $R_{0,3} = 0$ individuals.

We will assume that the proportion of time an individual spends at a certain location is governed by the mixing matrix, P :

$$P = \begin{bmatrix} 0.75 & 0.15 & 0.25 \\ 0.10 & 0.70 & 0.25 \\ 0.15 & 0.15 & 0.50 \end{bmatrix}.$$

Time (in days)	Number cases	Time	Number cases	Time	Number cases
1	2	11	53	21	13
2	5	12	55	22	14
3	10	13	58	23	11
4	12	14	53	24	12
5	14	15	52	25	9
6	15	16	42	26	7
7	25	17	40	27	5
8	31	18	30	28	4
9	42	19	23	29	2
10	45	20	19	30	1
-	-	-	-	31	1

Table 3: Influenza in an English boarding school^[1]

1. Write the system of equations that describe the dynamics of seasonal influenza in Figure 1 using standard incidence.
2. Following the directions in the R script *Case Study Seasonal Influenza*, plot the solutions $E_i(t)$ and $I_i(t)$ where $i = 1, 2, 3$. for $t = 150$ days using the initial conditions listed below Table 2.
3. We would like to improve the parameter values from the given initial conditions. Thus, consider a single patch with the same initial values except for $S_0 = 578$. Estimate the parameters β , σ , γ , and μ using the case data (i.e. I_t) in Table 3.
4. In order to parameterize the per capita movement rate between patches, we need to convert the mixing matrix into an adjacency matrix. Estimate the adjacency matrix from the cell phone data using the R script *Case Study Seasonal Influenza*.
5. Suppose that the first column of the adjacency matrix from part 4 represents Accomack county, the second column represents Fairfax county, and the third column denotes Montgomery. Use the adjacency matrix to generate a network describing the movement of individuals between these three counties.

6. Up to this point, we have been using the same parameters for all of the patches. What happens to the infectious and infected compartments if you change the parameter values, mixing matrix, and/or initial conditions? Describe your observations.
7. The number of secondary cases an infectious individual can produce in a population of fully susceptible individuals is known as the basic reproduction number, $\mathcal{R}_0$. For a specific patch, the basic reproduction number is $\mathcal{R}_{0,i} = \frac{\beta_i \sigma_i}{(\sigma_i + \mu_i)(\gamma_i + \mu_i)}$. Using the R script *Case Study Seasonal Influenza*, compute the patch specific reproduction numbers $\mathcal{R}_{0,i}$ and the network basic reproduction number for this case study.
8. When there is a seasonal influenza outbreak, there are several control measures that you can implement to reduce the transmission of the disease. For example, you can stay home and not interact with other people. What are other control measures that you can implement to reduce disease transmission?
9. While identifying which control measures mitigate seasonal influenza transmission is important, a more precise approach requires quantifying how specific system dynamics respond to variations across different parameter regimes. *Elasticity* provides the percentage change in the quantity Q with respect to the percentage change in the parameter p . Mathematically, we define *elasticity* as

$$\epsilon_p^Q = \frac{\partial Q}{\partial p} \frac{p}{Q}.$$

Therefore, if $\epsilon_p^Q = -0.5$, this means that a 1% increase in the parameter p will produce a 0.5% decrease in the quantity Q . Compute $\frac{\partial \mathcal{R}_{0,i}}{\partial p} \frac{p}{\mathcal{R}_{0,i}}$, where p represent the parameters listed in Table 2.

10. Using part 9), determine which control measures most effectively influence $\mathcal{R}_{0,i}$ for seasonal influenza.

References

- [1] Martcheva, Maia. *An introduction to mathematical epidemiology*. Vol. 61. New York: Springer, 2015.